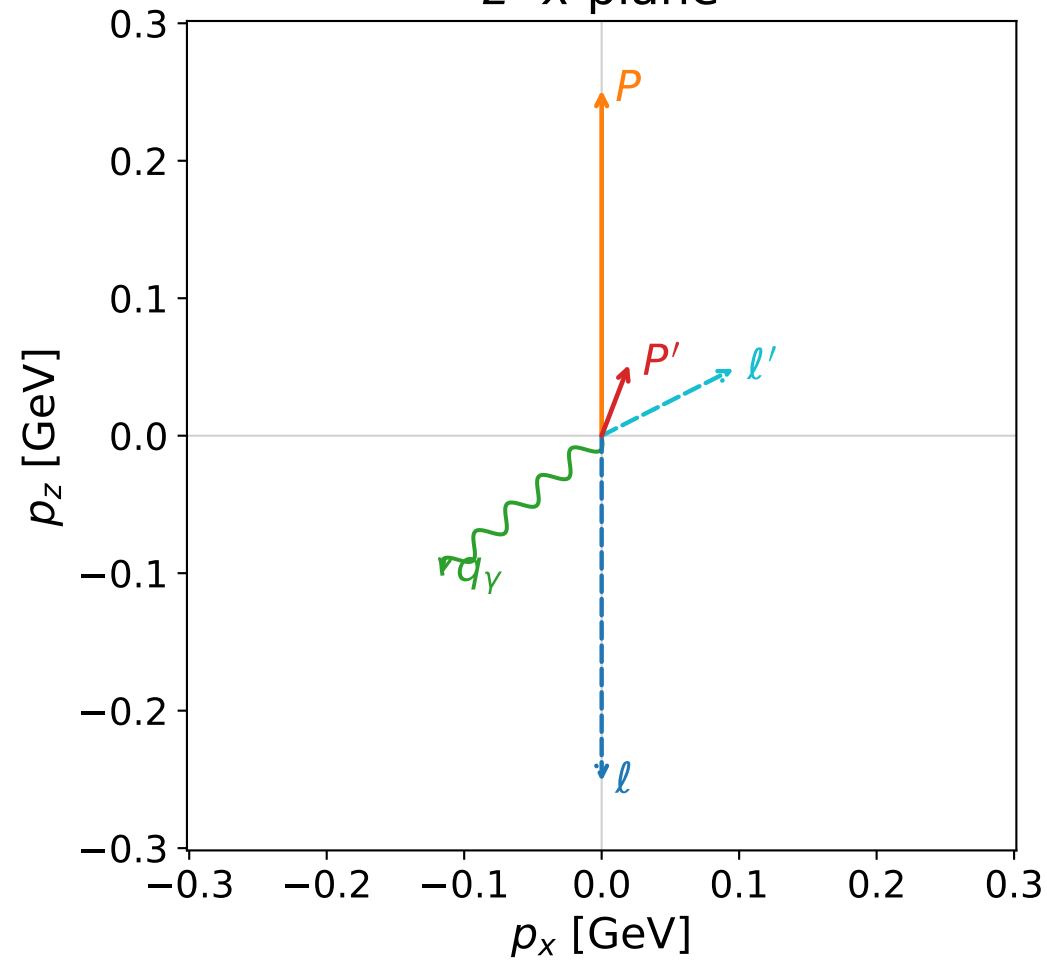


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

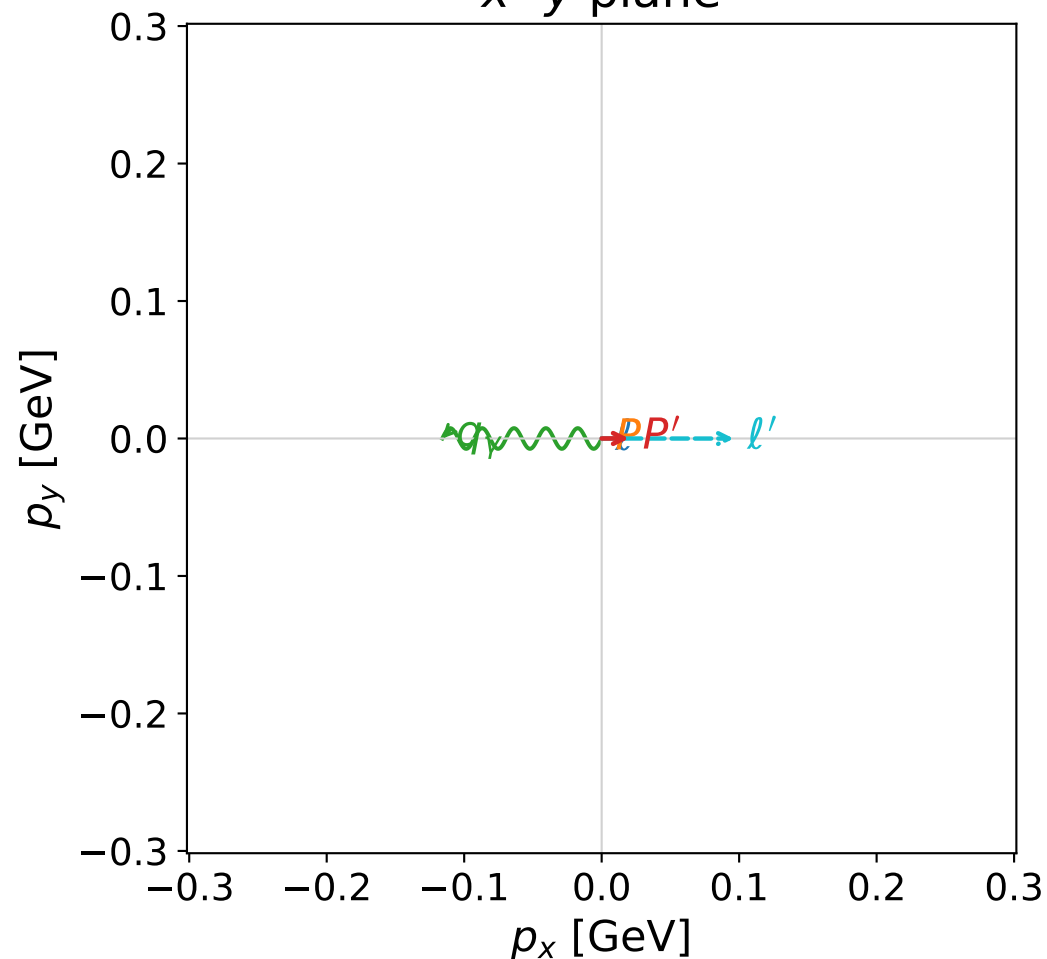


dw_mixing_angles_polarization_04_local_minimum_4
 $D_W=1.93491\text{e-}05$
 region: polarization_cluster_04_local_minimum_4
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.4990354$ rad
 incoming lepton: $0.0716994|+\rangle + 0.997426|-\rangle$
 proton mixing angle: $\alpha_p=1.8615242$ rad
 incoming proton: $-0.28665|+\rangle + 0.958035|-\rangle$

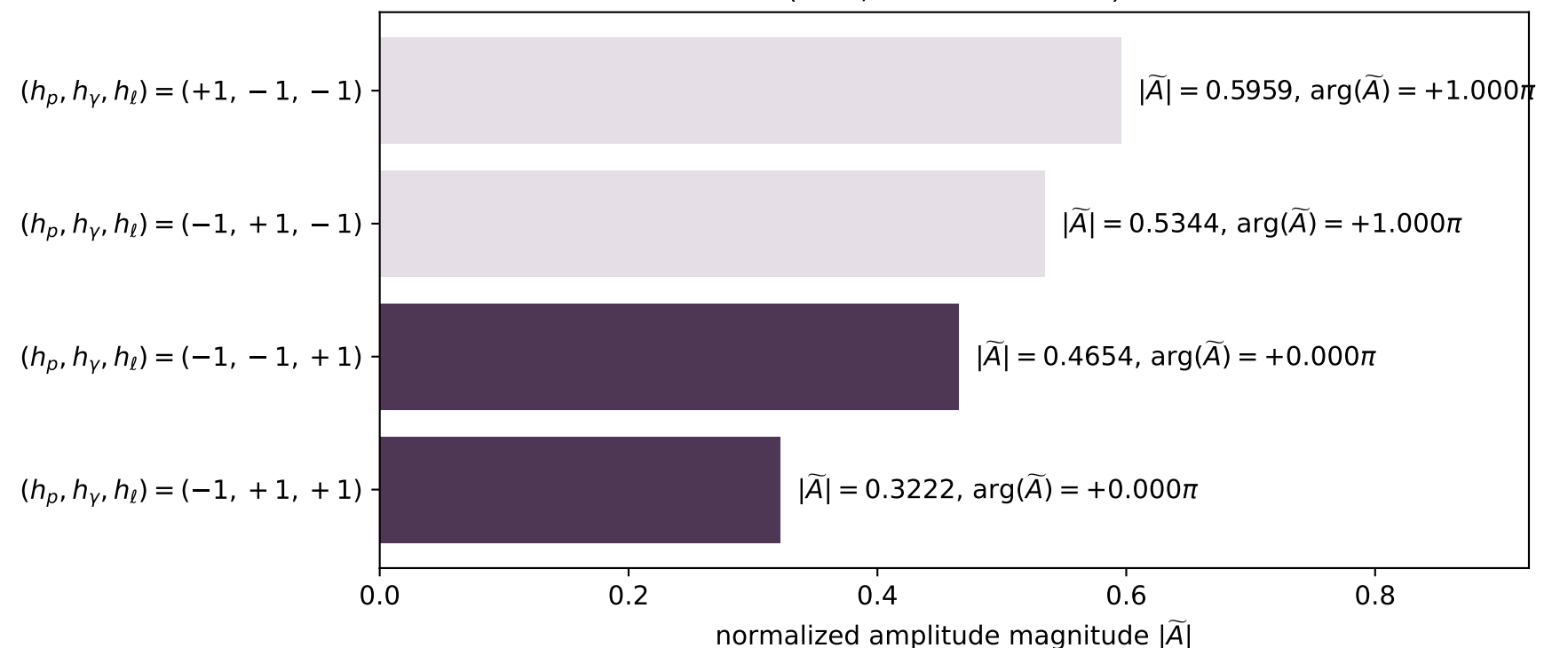
$s=1.54699$, $\sqrt{s}=1.24378$, $|\vec{P}|=0.25138$, $|\vec{P}'|=0.055368$
 $\theta_{p'}=0.367252$, $\theta_\gamma=2.28404$, $\phi_{p'}=0.00286392$, $\phi_\gamma=3.14164$
 $E_\gamma=0.153302$, $\theta_{l'}=1.10215$, $\phi_{l'}=6.28265$
 $Q^2=0.0843874$, $x_B=0.217797$, $t=-0.0392865$, $W^2=1.18292$, $y=0.590652$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.2727$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2514$
 P $E= 0.9711$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2514$
 l' $E= 0.1508$ $p_x= 0.0961$ $p_y= -0.0001$ $p_z= 0.0486$
 P' $E= 0.9396$ $p_x= 0.0199$ $p_y= 0.0001$ $p_z= 0.0517$
 q_γ $E= 0.1533$ $p_x= -0.1159$ $p_y= -0.0000$ $p_z= -0.1003$

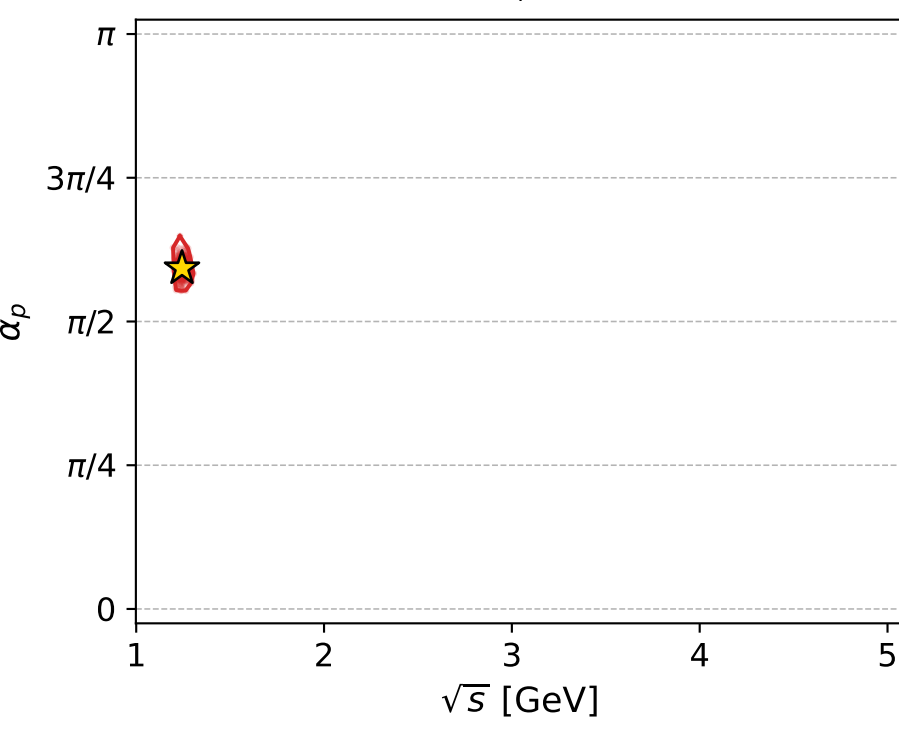
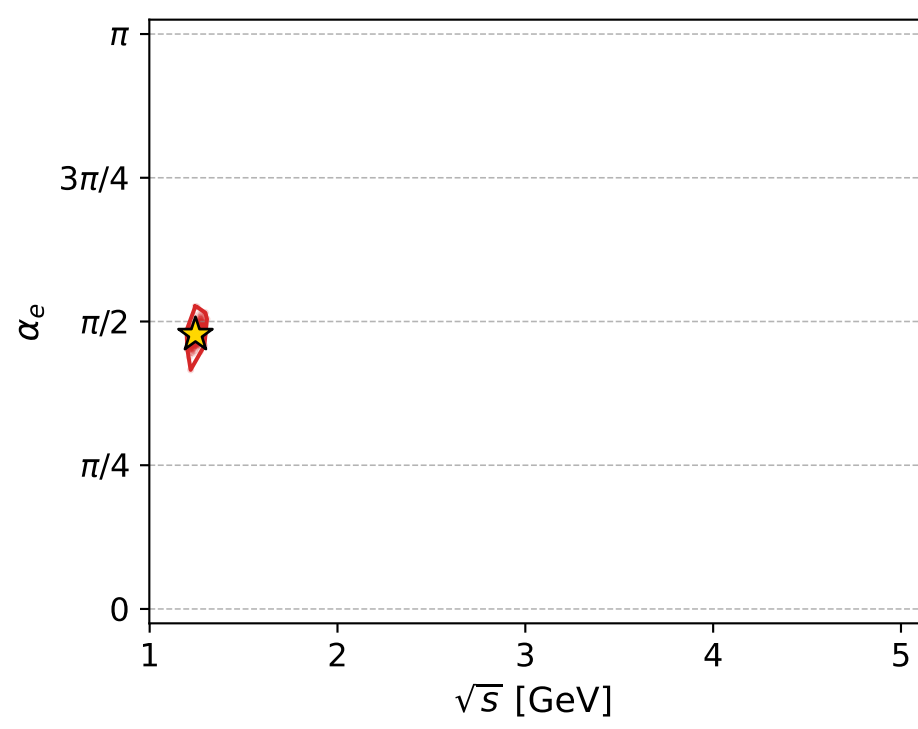
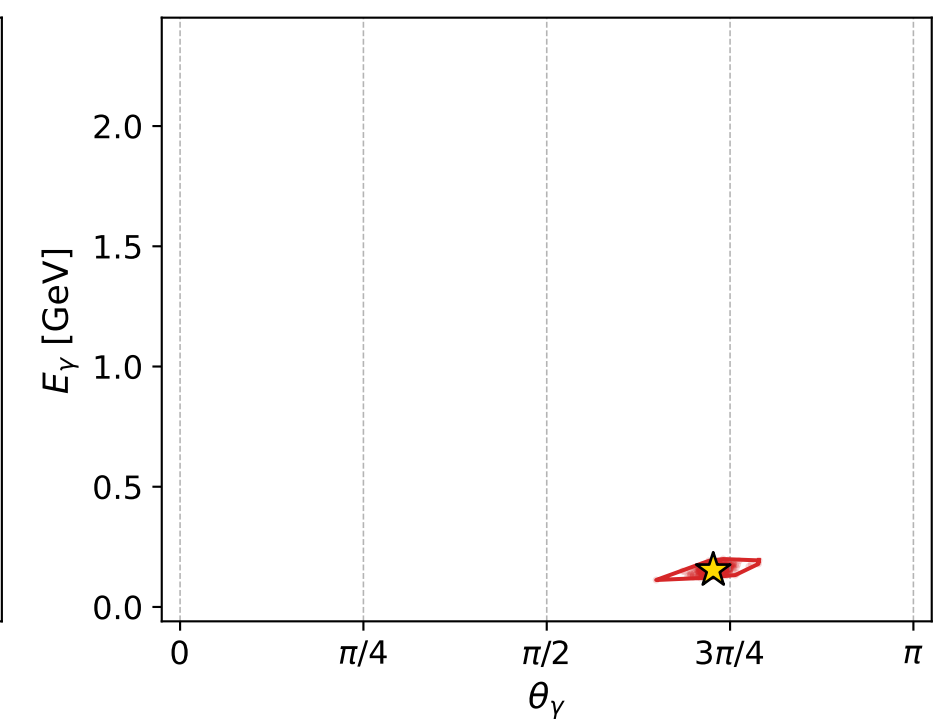
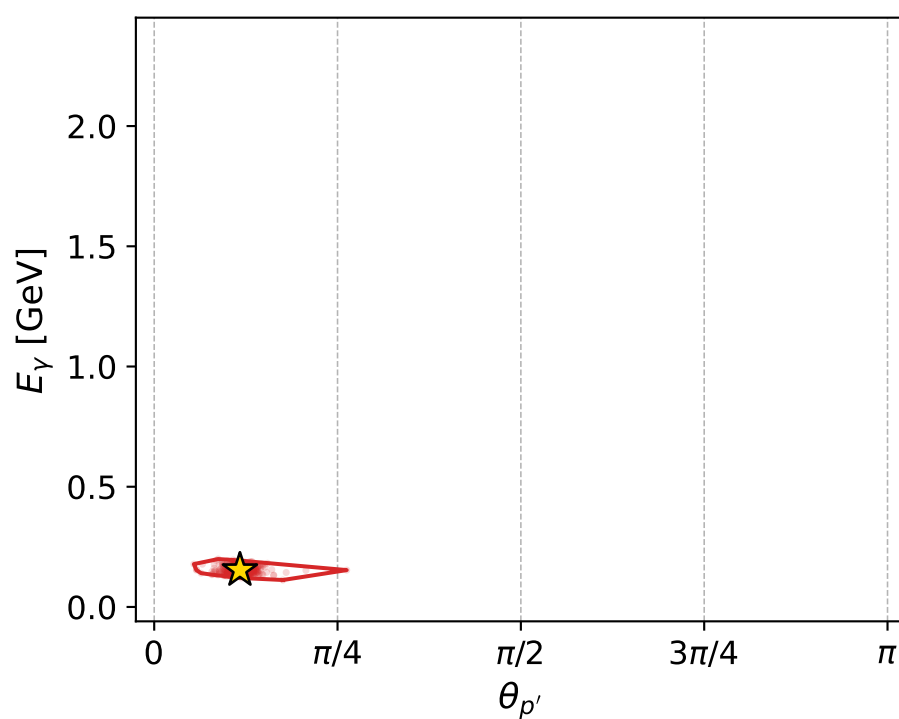
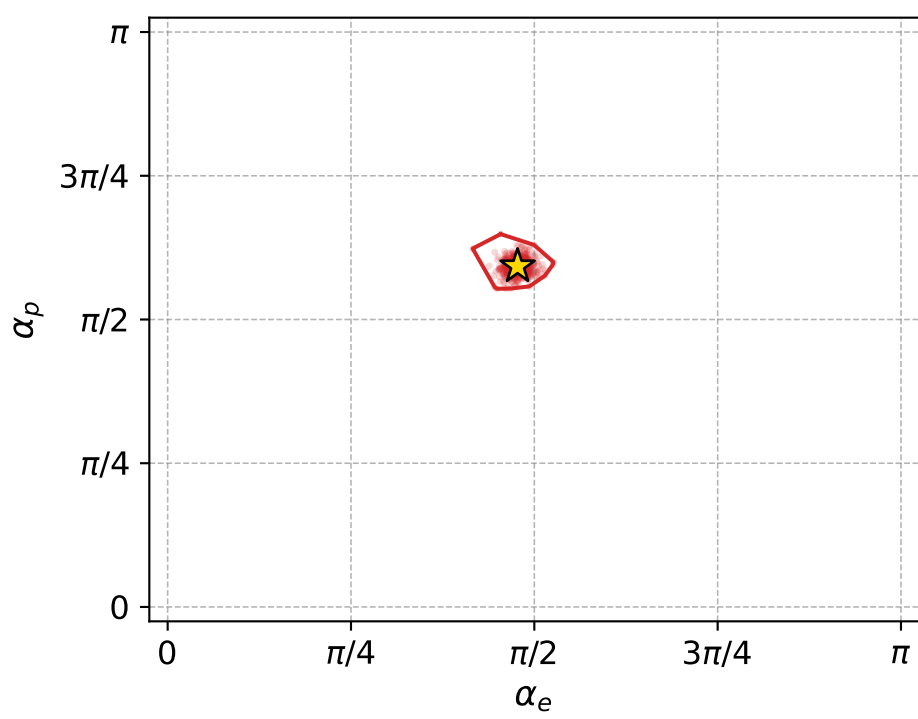
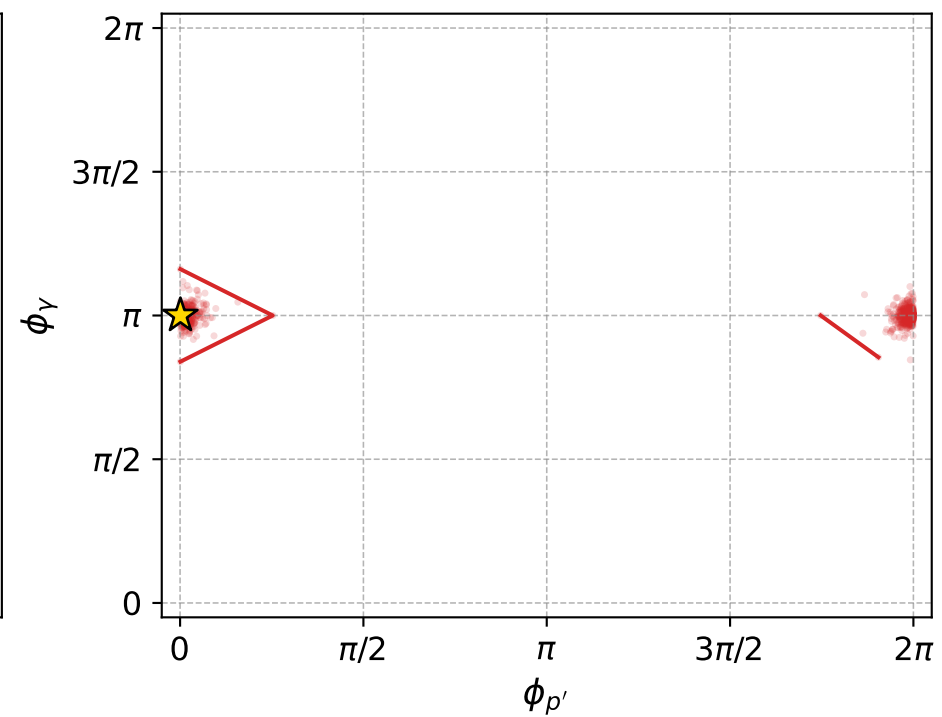
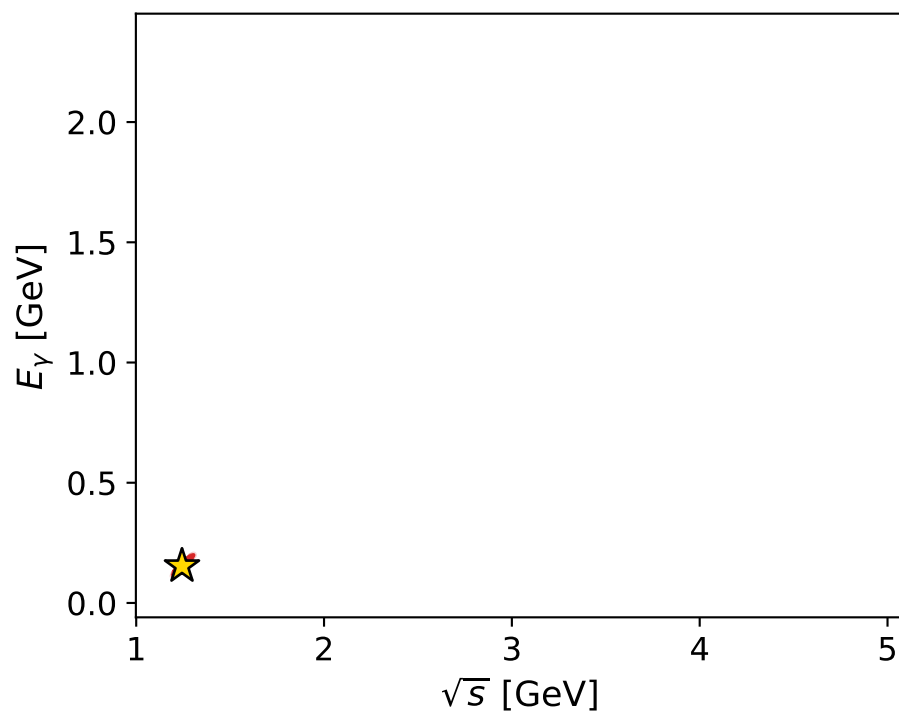
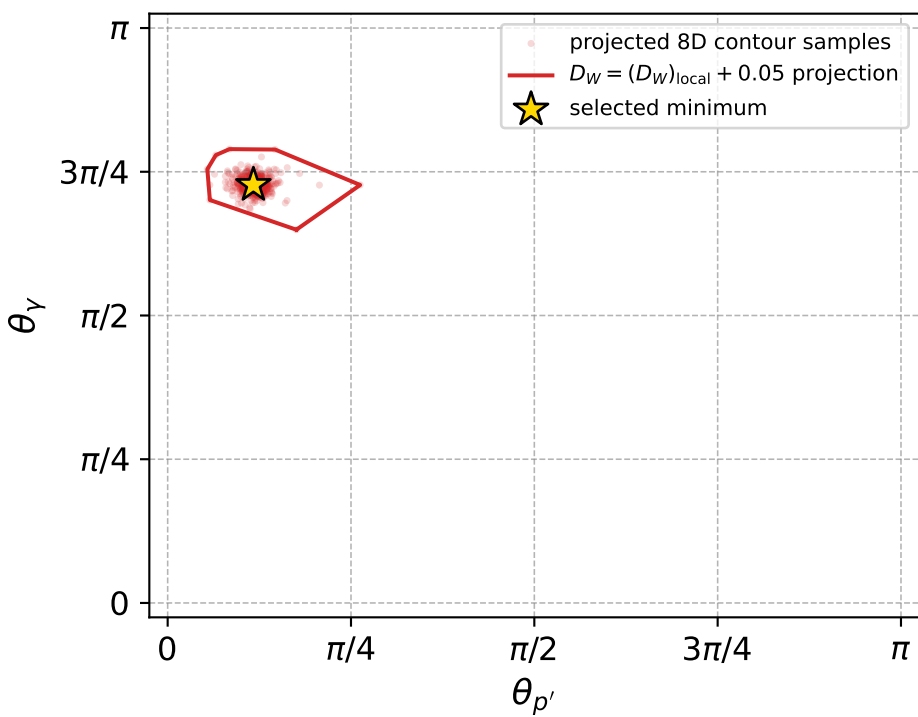
x--y plane



Leading normalized final-state amplitudes
 (N=4, retained=96.1%)



muon: polarization cluster P4, local minimum 4; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 1.9349101\text{e-}05$
 D_W contour = 0.050019349
 above global minimum = 1.5816315e-05
 8D contour samples = 1536
 configured 8D directions = 1536

selected phase-space point:

$\text{sqrt_s} = 1.243782$
 $\text{theta_p_out} = 0.3672523$
 $\text{theta_gamma_out} = 2.284039$
 $\text{q0ut} = 0.1533024$
 $\text{phi_p_out} = 0.002863916$
 $\text{phi_gamma_out} = 3.141642$
 $\text{alpha_e} = 1.499035$
 $\text{alpha_p} = 1.861524$